

ASJ UNIFIED BUSINESS SYSTEM

Software Requirements and End-to-End Workflow Specification

Prepared for	ASJ dry-fruits, gifting and box operations
Audience	Software developer, database engineer and application designer
Version	1.4 - product-group, recipe and quantity model
Date	28 August 2026
Build basis	This approved specification plus controlled owner-provided data imports
Primary objective	Build one unified inventory, costing, sales and reporting application

BUILD PRINCIPLE: One business event must be entered once. The system then updates inventory, costing, profit, customer history and reports automatically through the same permanent transaction ID.

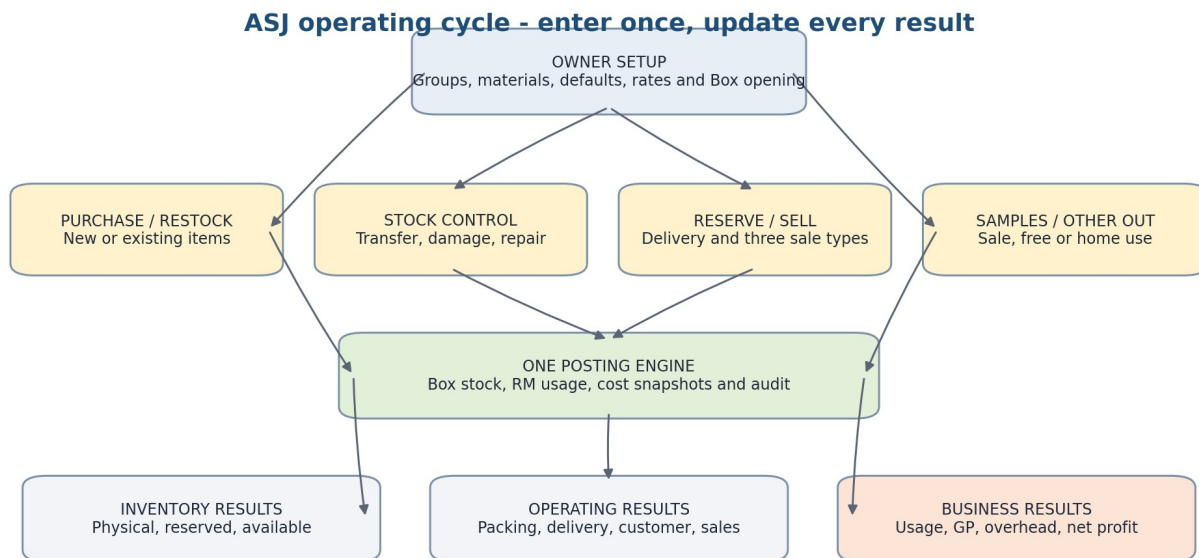
NON-NEGOTIABLE: No hard deletion of transactions. Corrections create versions or reversal entries. Historical sales, cancelled records, images and cost snapshots must remain traceable.

FIVE-DAY SPRINT: Target an integrated, mobile-ready MVP using temporary TEST data. Production use begins only after the acceptance matrix, security, backup/restore and opening-balance reconciliation pass; failed tests block go-live.

DATA ACCESS: The current ASJ Costing Sheet is incomplete and is not a build source. Do not request, import, connect or copy it. Controlled Inventory data is supplied later only after the import structure is approved.

EXECUTIVE WORKFLOW - WHAT THE SYSTEM DOES

AUTHORITATIVE BUILD BASIS: Build from this specification. The incomplete ASJ Costing Sheet is not imported, connected or treated as a source of truth. The owner provides controlled production data only after the application structure and acceptance tests are approved.



0.1 End-to-end business sequence

1. **Configure masters.** Create permanent Box, Raw Material, Product Group, Godown, Customer, Supplier, Gift Bag and user identities; set units, active dates and permissions.
2. **Configure material selection.** Create Raw Materials under Product Groups, set optional default quantities and optionally save common Box Default Recipes. Every active material remains addable to every box.
3. **Load the opening position.** Import owner-approved box quantities by godown and reconcile Physical, Reserved and Available Box Stock. Import Raw Material definitions and their two effective-dated rates, but no raw-material stock quantity.
4. **Record box purchases and restocking.** A new box design creates a permanent ID; an existing design reuses its ID. Accepted box quantities add stock and approved box costs create effective-dated rate periods.
5. **Control inventory.** Transfers move stock between godowns; repairable damage moves stock to Repair Hold; repair restores it; irreparable damage and other stock-out reduce stock and create the correct expense.
6. **Calculate or reserve.** Price Calculator gives a non-posting quotation. Reservations hold physical boxes only; raw-material lines are packing requirements because raw-material stock is not maintained.
7. **Create the sale.** BOX ONLY, BOX WITH ITEMS and ITEMS COMBO - NO BOX may appear as multiple lines under one OrderID. Actual Packing Quantity, not a preset, drives usage and staff instructions.
8. **Handle samples.** Saving a Sample recipe or changing Units Made creates no stock movement. Transfer to Sale creates a normal order; free/home/personal use creates one stock-out and expense.
9. **Post all results together.** The transaction engine writes Box Stock movements, Raw Material Usage, rate/cost snapshots, customer links, audit events and report totals together; a failed action writes nothing.

10. Review and correct safely. Registers and dashboards show stock, packing, sales, usage and profit. Edit/Update posts only the net difference; Cancel/Reverse preserves history and reverses effects once.

0.2 Current business function to future application result

Business function	Future application workflow	Result produced
Box Inventory	Box Master + Stock Dashboard	Permanent image/code and Physical, Reserved, Available and Repair Hold stock by godown
Stock Movement	Purchase, Transfer, Damage/Repair and Other Stock Out	One permanent movement ledger with source document, location, quantity, user and reversal history
Product / Raw Material Setup	Product Group -> exact Raw Material -> Unit/default quantity/two rates	Group-filtered selection, active/inactive control and optional Box Default Recipe without RM stock
Price Calculator	Non-posting quotation	Recommended price from selected calculator quantity, rates effective on Calculation Date and markup; no stock or sale record
Reservation Entry	Reservation + Delivery workflow	Reserved box stock, finalized rate, delivery details, pending packing-detail alert and conversion to sale without double deduction
Sales Entry	Multi-line Sales Order	Box Stock deduction when applicable, rate snapshots, revenue, customer history, Packing Sheet, calculated raw-material usage and both GP views
Sales Register	Searchable order register	Active/cancelled history with Edit, Update, Cancel, quantities, status, box/sample indicator and totals
Sample Combo	Saved Sample recipe	Actual per-unit packing quantities, Calculation-Date price and Units Made counter; no stock movement
Sample Master	Sample disposition workflow	Transfer to Sale or Give/Home/Personal Use, with Box Stock when applicable and calculated Raw Material Usage posted exactly once
Raw Material Usage	Calculated usage ledger/report	Monthly/yearly KG or PCS consumption calculated from Actual Packing Quantity; it is not a physical stock balance
Total Profit	Profit Dashboard	Revenue, boxes sold, combo sets sold, Commercial GP, Actual GP, overhead, stock-out expense and Net Profit

0.3 Results the owner and staff must receive

Result area	Mandatory output
Inventory control	Saleable, Repair Hold, Total Physical, Reserved, Available and Written Off for Boxes by godown; optional Gift Bag stock only when enabled
Packing execution	Searchable Packing Sheet showing box image, box/set count, Actual Packing Quantity per unit and Total Required for every material

Result area	Mandatory output
Customer and delivery	Reusable Customer Master, optional phone, order/reservation history, promised delivery details and readiness
Sales control	Sales Register with permanent OrderID, status, multiple lines, box/sample indicator, quantities, selling rate and revenue
Cost and profit	Applied rate snapshots, purchase-based Actual GP, internal-rate Commercial GP, margins and realized markup
Management reporting	Calculated raw-material usage, box purchases/damage, boxes sold, combo sets sold, fixed overhead, stock-out expense and Net Profit by month/year
Control and recovery	Role access, version history, reversals, audit log, encrypted backups, tested restore and controlled TEST-data reset

0.4 Data access and implementation boundary

- **No Costing Sheet access.** The current Costing Sheet is incomplete. The developer must not request access, import its rows, synchronize it or copy its formulas.
- **Specification is authoritative.** When a current Sheet and this document differ, this approved document controls the new application.
- **Inventory data comes later.** Live Inventory access is not required to start. After the developer supplies the final import template, the owner may provide a controlled export or temporary read-only access for Box images/codes and opening quantities.
- **Temporary data first.** Every workflow is built and tested using clearly labelled TEST customers, boxes, materials, rates, orders and movements.
- **Production handoff only after acceptance.** The owner supplies final opening Box Stock by godown, box costs/images, Raw Material definitions and two rates, and settings only after mobile/desktop testing, backup/restore and reconciliation pass.

0.5 Simple examples - owner action to final result

Build each process using the same simple rule: the owner/staff enters one business event, and the application creates every connected result automatically. These examples explain the intended behavior before the technical sections.

A. Setup and Box Inventory examples

Process	Simple ASJ example and required result
Opening setup	Owner enters ASJ001 opening Box Stock: Home 100 and Shop 20. Result: Physical 120, Reserved 0, Available 120. No Raw Material stock is entered.
Raw Material setup	Owner creates ALMOND PREMIUM, Unit KG, Purchase/Actual Cost Rate INR 800/KG and Internal Rate INR 900/KG effective 01/09/2026. Result: the rates are available for costing; Almond stock remains untracked.
Product Group setup	Owner creates Product Group CASHEW, then adds CASHEW W320 and CASHEW W240 under it. Result: selecting CASHEW shows only its active variants, while reports can total the complete Cashew group.

Process	Simple ASJ example and required result
Raw Material quantity setup	Owner gives CASHEW W320 a default quantity of 0.250 KG. Result: 0.250 prefills when selected; the user may overwrite it. If no default exists, the quantity stays blank for manual entry.
New box purchase	Owner buys 50 new boxes at INR 100 each with INR 500 transport. Transport per box = INR 10; landed cost = INR 110. Result: create BoxID/code/image, add 50 to the selected godown and start the box cost period.
Existing box restock	Owner selects the existing ASJ001 image and buys 30 more. Result: reuse the same BoxID, add 30 stock and never create a duplicate design.
Repairable damage	Five ASJ001 boxes are damaged but repairable. Result: move 5 from Saleable to Repair Hold; Total Physical is unchanged and Available falls by 5.
Repair / write-off	Three of those five boxes are repaired and two cannot be repaired. Result: 3 return to Saleable; 2 leave stock. At INR 110 loss each, INR 220 posts once to STOCK OUT expense.
Godown transfer	Move 20 ASJ001 from Home to Shop. Result: Home -20, Shop +20 and total company Physical Stock is unchanged.
Reservation	Customer reserves 10 ASJ001 boxes. Result: Physical is unchanged, Reserved +10 and Available -10. A recipe such as 0.250 KG Almond per box may be saved, but no Almond stock is reserved.
Partial delivery	From those 10 reserved boxes, customer takes 6 today. Result: create one sale for 6, reduce Physical and Reserved Box Stock by 6, calculate RM Usage for 6 and keep the remaining 4 boxes reserved.

B. Sales, costing and sample examples

Process	Simple ASJ example and required result
BOX ONLY sale	Sell 5 ASJ001 boxes. Result: Box Stock -5, Boxes Sold +5, revenue/cost snapshots created and Raw Material Usage = 0.
BOX WITH ITEMS	Sell 2 boxes containing 0.250 KG Almond each. Result: Box Stock -2 and calculated Almond Usage = 0.500 KG. No Almond stock balance is reduced.
Box recipe and Add Material	ASJ001 normally uses Almond 0.250 KG and Cashew 0.250 KG, so those lines prefill from its optional recipe. For one order, owner taps ADD RAW MATERIAL, selects Product Group RAISIN -> RAISIN PREMIUM and enters 0.100 KG. Result: all three actual quantities are saved for that order; the permanent ASJ001 recipe changes only if the owner separately chooses SAVE AS DEFAULT RECIPE.
NO-BOX combo	Sell 3 sets containing 0.250 KG Almond each. Result: no Box Stock movement, Combo Sets Sold +3 and calculated Almond Usage = 0.750 KG.
Preset override	Preset shows 0.250 KG, but owner changes Actual Packing Quantity to 0.300 KG for 4 boxes. Result: Packing Sheet and Usage both use 1.200 KG, not the old preset.
Raw Material costing	Billing Date rate is Purchase INR 800/KG and Internal INR 900/KG. For 0.250 KG, Actual Cost = INR 200 and Commercial Cost = INR 225. FIFO is not used.

Process	Simple ASJ example and required result
Rate change	Almond Purchase Rate changes from INR 800 to INR 820 effective 15/09/2026. Result: sales before 15 September keep INR 800; later sales use INR 820. Old history is not overwritten.
Active / inactive	Almond Premium becomes inactive from 01/10/2026. Result: it is hidden from new October entries but remains visible in every old order and report.
Price Calculator	Owner selects a box and materials to see a recommended rate. Result: price only; no customer, sale, reservation, Box Stock or Raw Material Usage record.
Customer	Staff types MAYANK. If the customer exists, select the saved CustomerID; otherwise create a new customer with name only because phone is optional. Result: future sales can reuse the same customer and history.
Packaging	Packing costs INR 20 and customer separately pays INR 10. Result: revenue includes the INR 10 once, direct cost includes INR 20 once and Net Packaging Burden displays INR 10.
Sample recipe	Owner saves a sample recipe and Units Made = 10. Result: no stock or usage. If 2 units are transferred to Sales, create one normal sale and usage. If 1 is given free, create usage and STOCK OUT expense once.
Edit / Update	An order changes from 2 boxes to 3. Result: keep the same OrderID and post only the extra 1 box plus the extra calculated Raw Material Usage.
Cancel	A posted order is cancelled. Result: reverse its Box Stock effect once, exclude its active Raw Material Usage, revenue and GP, and keep the cancelled history.
Monthly report	In September, 40 physical boxes and 12 no-box combo sets are sold. Result: Boxes Sold = 40 and Combo Sets Sold = 12; do not combine them.

DEFINITION OF DONE: One business entry must update every relevant stock, costing, packing, customer, usage and profit result without a second manual entry, while remaining reversible and auditable.

How to use this specification

This is the functional contract for the developer. Read the Executive Workflow first to understand the complete operating cycle and the results expected from every entry, then use Parts A-G for detailed field, calculation, database, security and acceptance requirements. The approved rules in this document are authoritative; incomplete current Sheets are not implementation specifications.

DEVELOPER HANDOFF: Build the core as a single transactional application. Google Sheets may be used only for controlled future imports/exports or management reports; the incomplete Costing Sheet must not be accessed, imported or used as the primary database/calculation engine.

Document map

Part	Coverage
Executive workflow	Complete business cycle, current-function mapping, automatic outputs and data-access boundary
A. Foundation	Objectives, principles, roles, architecture and master data
B. Inventory	Box opening stock, box purchase, damage, transfer and movement rules
C. Costing and sales	Reservation, effective rates, three sale types, gift bags, updates and cancellation
D. Pricing and samples	Price Calculator, Sample Combo, physical sample and conversion to sale
E. Reporting	Sales Register, raw-material usage, profit, customer and stock reports
F. Technical delivery	Database, APIs, security, audit, tests, migration and rollout
G. Acceptance tests	End-to-end pass conditions and production-readiness gate

Decisions already fixed

- **One unified system.** Inventory and costing will share the same database and transaction service.
- **Temporary data first.** The complete system is tested before production data is entered.
- **Permanent history.** Cancelled/inactive records stay visible for audit and are excluded from active calculations.
- **Billing Date drives cost.** Applicable box, raw-material and gift-bag rates are determined by the transaction Billing Date.
- **Two raw-material rates.** Purchase / Actual Cost Rate is used for actual profitability; Internal Selling/Costing Rate is used for pricing and commercial profitability.
- **Raw materials are non-stock in this scope.** The owner maintains prices, recipes and calculated usage only. The application does not ask for raw-material opening stock, purchase quantities, godowns, reservations, transfers or FIFO layers.
- **Overhead stays separate.** Rent, salary, electricity, software and administration are never included in box or raw-material direct cost.
- **Images lead selection.** Users select boxes through image cards; Box Code is displayed and stored automatically.
- **Units.** Raw materials use KG or PCS. KG accepts decimals such as 0.250; PCS must be a positive whole number.
- **Two quantity concepts.** An optional Selected Quantity preset may prefill entry, but the editable Actual Packing Quantity is the saved source of truth for packaging, samples, sales and usage.
- **Raw materials remain flexible by box.** Every active Raw Material is eligible for every Box by default. An optional Box Recipe Template may prefill common items, but users can add any other active material through Product Group -> Raw Material selection.

PART A - FOUNDATION

1. Product outcome

The application will manage ASJ box inventory, godowns, box purchases, raw-material rate masters and calculated usage, reservations, sales, costing, sample recipes, delivery follow-up and profitability from one controlled interface. A user enters each business event once; the backend posts every required ledger and report automatically.

- Responsive web/mobile app suitable for owner and staff use, with no desktop-only daily workflow.
- Owner-only backend screens for rates, cost rules, overhead and corrections.
- Image-first box selection with unique permanent identity.
- Atomic transaction posting: all related records succeed together or none are written.
- Fast monthly/yearly reports without recalculating the complete database on every screen load.
- Exports to Google Sheets/Excel/PDF where required; Tally remains the billing/accounting platform unless separately integrated.

2. Non-negotiable data principles

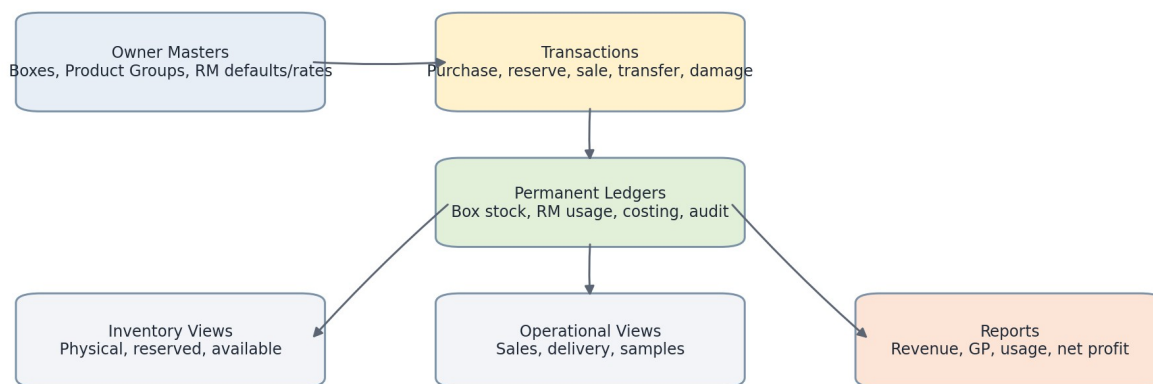
Principle	Required behavior
Permanent internal IDs	Every box, material, customer, supplier, reservation, order and movement has an immutable ID. Display codes/names are attributes, not relational keys.
No hard deletion	Business transactions are cancelled/reversed. Masters become inactive. Historical links remain valid.
One source of truth	Balances are derived from ledgers. Screens and reports never maintain independent editable totals.
Snapshot + rate link	Transactions store applied Rate History ID, applied rate and calculated amount. Future rate updates do not silently alter old transactions.
Audited correction	A genuine historical-rate correction may recalculate affected open-period transactions, but must log before/after values, user, time and reason.
Idempotent posting	Retrying Save/Sync must not create duplicate movements, order lines or cost records.
Status filtering	ACTIVE/POSTED records drive stock and reports. CANCELLED records remain searchable but contribute zero.

3. User roles and permissions

Role	Allowed	Restricted
Owner / Super Admin	All masters, rates, effective dates, costs, overhead, corrections, cancellations, reports and user access	Nothing, but every correction is audited
Manager	Purchase, sale, reservation, transfer, damage approval, customer and operational reports	Cannot edit historical rate dates, delete records or change security
Staff / Operator	Create assigned entries, view active stock and delivery work	Cannot see owner-only purchase costs, internal rates, GP, overhead or backend audit
Viewer / Accountant	Read/export selected registers and reports	No transaction posting or master edits

4. Logical architecture

One database, multiple controlled workflows



- Recommended database: PostgreSQL or an equivalent relational database with transactions and constraints.
- Application layer: REST/GraphQL APIs or equivalent service layer; calculations must live in tested domain services, not UI formulas.
- Files/images: permanent object storage/Drive paths, with stable Box Image references in the database.
- Reporting: read-optimized queries/materialized summaries refreshed after successful posting, not before.
- Optional Sheet export: generated from database views; never used to overwrite the core ledger.

5. Core master data

5.1 Box Master

Field	Rule
BoxID	System UUID/sequence; immutable primary key
BoxCode	Human code such as ASJ001; globally unique across active/inactive/history; cannot be reused

Field	Rule
BoxImage	Permanent file reference; required for active sale selection
DisplayOrder	Owner-controlled sort number. Rearranging images changes this field, never BoxCode
BoxName / Category / Size	Optional descriptive fields for search and grouping
Active	Owner-controlled. Inactive boxes remain in history and purchase/restock search
AvailableFrom / InactiveFrom	Controls date-based eligibility where required
Selection behavior	New-entry screens show active boxes with Available Stock > 0. Purchase can select any existing BoxID to restock

IMPORTANT: After production begins, BoxCode is permanent. To change visual arrangement, use DisplayOrder. A box at zero stock becomes unavailable for new sales but its historical orders and images remain unchanged.

5.2 Godown Master

- Initial godowns: HOME GODOWN ROOM 1 and SHOP.
- GodownID is permanent; name may be changed by the owner.
- Each stock movement has Source Godown and/or Destination Godown as applicable.
- New godowns can be added later without schema changes.

5.3 Raw Material Master

Field	Rule
RawMaterialID	Permanent internal ID
Product Group	ALMOND, CASHEW, RAISIN, etc.; managed in Product Group Master
Variant / Type	Optional detail such as W320, Premium or Regular
Display Name	Generated or owner-approved combination; unique among active materials
Unit	KG or PCS only; immutable after transactions exist
Available From	Material appears only for Billing Dates on/after this date
Inactive From	If set, material is unavailable for new transactions on/after this date
Status	ACTIVE by default; INACTIVE hides it from new selections but preserves history
Default Selected Quantity	Optional quick-entry value in the material's Unit. It prefills entry but never prevents manual overwrite

Field	Rule
Current Purchase / Actual Cost Rate	Read-only current value from Rate History: owner-provided cost per KG/PCS used for Actual Gross Profit; no stock quantity or FIFO layer is inferred from this rate
Current Internal Selling/Costing Rate	Read-only current value from Rate History: owner-defined pricing value used in Price Calculator, Sample Combo and Commercial GP

5.3.1 Product Group and Raw Material creation

- Product Group is the broad family (for example ALMOND or CASHEW); RawMaterialID is the exact costed/packed variant. Group alone is never used as an order item.
- Create the Product Group first with GroupID, Name, Available From, optional Inactive From and Status. Then create each exact Raw Material under one Product Group.
- Example: Product Group CASHEW contains CASHEW W320 and CASHEW W240. The transaction saves W320 or W240, while reports may total both under CASHEW.
- Entry search works in two stages: select/search Product Group, then show only active Raw Materials belonging to that group for the transaction date.
- Inactivating a Product Group hides all its materials from new entries. Inactivating one Raw Material hides only that variant. Old recipes, orders, rates, usage and reports remain readable.
- A newly created active Raw Material immediately becomes available through ADD RAW MATERIAL for every eligible box; it is not automatically inserted into old saved transactions or recipes.
- Unit becomes immutable after a material is used. A wrong-unit material is inactivated and replaced or corrected through an owner-only audited migration.

CREATION EXAMPLE: Create Product Group CASHEW -> create CASHEW W320 -> choose Unit KG -> optional Default Selected Quantity 0.250 -> enter both effective-dated rates -> activate. A user selecting CASHEW then sees W320 and its 0.250 KG default; Cashew stock is never requested.

5.3.2 Box eligibility and optional Default Recipe

- Every active Raw Material is eligible for every active Box by default. Do not create a restrictive Box-to-Material permission list.
- A Box may have an optional owner-managed Default Recipe containing its commonly used Raw Materials and default quantities. This is a convenience template, not a stock bill of materials.
- When the Box is selected, its active Default Recipe lines prefill. The user may edit a quantity, remove a line or tap ADD RAW MATERIAL to choose any other active Product Group and Raw Material.
- Adding/removing a material in Price Calculator, Sale, Reservation or Sample changes only that working document. The permanent Box Default Recipe changes only through an owner-only SAVE/UPDATE DEFAULT RECIPE action.
- If a Box has no Default Recipe, the item area starts blank and the user adds materials manually. If a recipe material later becomes inactive, keep it in historical documents but do not silently prefill it in a new transaction; flag the recipe for owner review.

BOX EXAMPLE: ASJ001 Default Recipe contains ALMOND PREMIUM 0.250 KG and CASHEW W320 0.250 KG. Those two lines prefill. For a special order, the owner adds Product Group RAISIN -> RAISIN PREMIUM 0.100 KG. The order saves three Actual Packing Quantities, but the ASJ001 Default Recipe stays unchanged unless the owner explicitly updates it.

5.3.3 Product Raw Material quantity concept

- Each material may have zero or more owner-managed Selected Quantity presets. Examples: 0.050, 0.100 or 0.250 for KG; 1, 2 or 4 for PCS.
- One preset may be marked Default. When the material is added, that value prefills automatically. The user may overwrite it. If no default/preset exists, the quantity remains blank and must be typed manually.
- The Box Default Recipe may store a box-specific prefill. Priority is: Box Recipe quantity -> Material Default Selected Quantity -> blank manual entry.
- Price Calculator uses the selected/editable calculator quantity only for that temporary calculation. Saved Sale, Reservation and Sample workflows copy the prefill into Actual Packing Quantity, which the user can overwrite and which becomes the permanent source of truth.
- KG is stored as decimal kilograms and rated in INR/KG: 0.250 means 250 g. The UI may additionally show '250 g' as a friendly display, but must save 0.250 KG. PCS is stored/rated per piece and rejects decimals.

NON-STOCK RAW MATERIAL POLICY: Raw Material Master stores identity, unit, active dates, quantity presets and two effective-dated rates. It does not store Physical, Reserved or Available Stock. Usage is calculated from saved Actual Packing Quantity when a sale or final sample disposition is posted.

RATE OWNERSHIP: Rates are never silently overwritten in Raw Material Master. The master may display the current value, but every change creates or corrects an effective-dated Rate History record with user, timestamp and reason.

5.4 Gift Bag Master

- Name is GIFT BAG for all rows.
- Sizes: SMALL, MEDIUM and LARGE.
- Each size has effective-dated Cost Rate and Use in Sales flag.
- If Bag Required = NO, Bag Size is hidden/disabled and quantity/cost are zero.
- If Bag Required = YES, only enabled sizes appear and size is mandatory.
- Bag quantity automatically equals the associated box/set quantity.
- Gift Bag is a direct cost in both Actual and Commercial GP. It never automatically changes a manually entered selling rate; the Price Calculator includes it when recommending a price.
- Core MVP treats Gift Bag as a costed consumable. Deduct Gift Bag stock only when an owner-controlled Track Inventory flag is enabled and opening/purchase quantities are maintained.
- When Track Inventory is enabled, Gift Bags behave as PCS stock items for opening, purchase, godown transfer, damage, sale/update/cancel and availability checks. When disabled, only their effective-dated cost is used and no stock movement is posted.

5.5 Other masters

Master	Minimum fields / purpose
Product Group	GroupID, Name, active dates/status; controls two-stage Raw Material selection, not box restrictions
Box Default Recipe	Optional BoxID template of common RawMaterialIDs and prefill quantities; every other active material remains addable
Supplier	SupplierID, name, phone, tax details, payment terms, active

Master	Minimum fields / purpose
Customer	CustomerID, name, optional phone, addresses and delivery notes; phone is never required or used as the primary key
Direct Cost Type	Packing, loading, special delivery or another order-attributable cost
Overhead Category	Godown rent, salaries, electricity, software/admin and other fixed/period costs
Business Settings	Markup, packing slabs, tax defaults, numbering prefixes, unit options and feature flags

PART B - INVENTORY AND PURCHASE

6. Opening data and production cutover

At production cutover the owner provides approved Box opening quantities by godown, permanent box images/codes and initial box costs. For Raw Materials, the owner provides definitions, units, quantity presets and the two effective-dated rates only. No raw-material opening quantity or physical balance is imported.

- 1. Import approved masters.** Validate Box Codes/images and Raw Material groups, variants, units, active dates and permanent IDs.
- 2. Import Box Opening Stock by Godown.** Create one POSTED OPENING_BALANCE movement per BoxID + GodownID.
- 3. Import opening rates.** Create the first effective Box Cost periods and Raw Material Purchase/Actual Cost and Internal Selling/Costing Rate periods using the agreed cutover date.
- 4. Reconcile.** Owner signs off physical, reserved and available Box totals before users can post production transactions. Raw Material validation checks rates and units, not stock quantity.
- 5. Lock cutover.** Opening entries can be corrected only by owner audit adjustment after production starts.

Total Physical Stock = Saleable Stock + Repair Hold Stock

Available Stock = Saleable Stock - Reserved Stock

Repairable damage remains physically owned in Repair Hold but is not available. Unrepairable damage leaves physical stock through a write-off movement.

7. Inventory ledger model

The app must not store a manually editable current balance. It stores immutable stock movements and calculates balances by ItemID and GodownID for Boxes and, only when enabled, Gift Bags. Raw Materials are excluded from the physical stock ledger; they use a separate calculated Usage Ledger. A business document may post several rows, but every row references the same Document ID.

Movement type	Physical effect	Reserved effect	Required reference
OPENING_BALANCE	+ quantity	0	Opening batch
PURCHASE_IN	+ accepted quantity	0	PurchaseID
TRANSFER_OUT	- quantity at source	0	TransferID
TRANSFER_IN	+ quantity at destination	0	TransferID
RESERVE	0	+ quantity	ReservationID
RESERVATION_SOLD	- quantity	- quantity	ReservationID + OrderID
RESERVATION_CANCEL	0	- quantity	ReservationID
DIRECT_SALE	- quantity	0	OrderID
DAMAGE_REPAIRABLE	0 total; Saleable -> Repair Hold	0	DamageID

Movement type	Physical effect	Reserved effect	Required reference
DAMAGE_REPAIRED	0 total; Repair Hold -> Saleable	0	Original DamageID
DAMAGE_UNREPAIRABLE	- quantity from Saleable/Repair Hold	0	DamageID + ExpenseID
DAMAGE_REVERSAL	Exact reverse of erroneous damage	as required	Original DamageID
SAMPLE_OR_HOME_OUT	- Box/Gift Bag quantity if used	0	SampleID + ExpenseID
OTHER_STOCK_OUT	- quantity	0	Reason + document ID
SALE_CANCEL_REVERSAL	+ prior sold quantity	as required	Cancelled OrderID
ADJUSTMENT	+/- quantity	0	Owner approval + reason

CONDITION BALANCE: For every stock-tracked Box/Gift Bag and godown, the ledger must distinguish Saleable, Repair Hold and Written Off. Only Saleable less Reserved is Available. Raw Materials are not part of this balance.

8. Purchase Entry - boxes

Purchase Entry is the only normal way to add new box stock after opening. It supports both a new design and an existing design.

8.1 Entry mode

Mode	User experience	Result
NEW BOX DESIGN	Upload image; enter unique BoxCode, name/category/size, supplier, quantity, godown and costs	Create BoxID, cost history and PURCHASE_IN movements
EXISTING BOX DESIGN	Select image card; BoxCode auto-fills; enter purchase details	Reuse same BoxID; add stock and start/update cost period

8.2 Purchase fields

- Purchase Date (mandatory), Supplier, Invoice/Reference, Destination Godown.
- Ordered Quantity, Accepted Quantity and Damaged/Rejected on Receipt.
- Unit Box Rate Before GST, GST %, Input Tax Credit (ITC) eligibility, recoverable/non-recoverable GST and purchase amount.
- Transportation Total and quantity over which transport is allocated.
- Other direct acquisition costs, if any. Fixed overhead is not allowed here.
- Unrecoverable damage provision or allocated receipt loss, where configured.
- Optional Special Extra Profit % for the specific box design/cost period.
- Notes, attachments and Created By.

RECEIPT DIFFERENCE: Supplier-rejected or returned quantity is not stock and is not a damage expense unless ASJ actually bears the loss. Only accepted quantity enters stock. Accepted-but-damaged quantity follows the Damage workflow with ownership, condition and cost impact recorded separately.

8.3 Purchase calculations

Transport per box = Transportation Total / Transport Allocation Quantity
Landed Unit Box Cost = Purchase Unit Cost + non-recoverable GST + Transport per box + Other Direct Cost per box + Damage Provision per box
Final Unit Box Rate = Landed Unit Box Cost x (1 + Special Extra Profit %)

Recoverable input GST is tracked separately and is not a product cost; only non-creditable GST enters Landed Unit Box Cost. If Special Extra Profit % is blank or zero, Final Unit Box Rate equals Landed Unit Box Cost. The percentage is not applied again in Price Calculator or Sample Combo; those tools consume the final rate.

8.4 Posting

1. Validate all fields and available master IDs.
2. Calculate landed and final pricing rates.
3. Create/close effective cost periods without overlap.
4. Post PURCHASE_IN stock movement for accepted quantity.
5. Post the applicable repairable/unrepairable damage or purchase-rejection record for unusable quantity, with remarks.
6. Commit all records in one database transaction and generate PurchaseID.

8.5 Raw Material rate update - no stock purchase entry

- Owner selects an existing RawMaterialID or creates a new material by first selecting/creating Product Group, then entering exact Variant/Name, KG or PCS, active dates and optional Default Selected Quantity.
- A newly activated material becomes addable to every Box through Product Group -> Raw Material. Inactive groups/materials are hidden from new entries but remain in all historical recipes, transactions and reports.
- Enter Effective From, Purchase / Actual Cost Rate per KG/PCS and Internal Selling/Costing Rate per KG/PCS. Supplier, invoice, godown, received quantity and rejected quantity are not required in the current scope.
- Saving a Raw Material rate creates or closes non-overlapping Rate History periods. It never posts PURCHASE_IN or creates a physical raw-material balance.
- Purchase / Actual Cost Rate is owner-provided and is used for Actual GP. The application does not calculate FIFO, weighted average or batch landed cost without quantity-based purchase records.
- Internal Selling/Costing Rate is entered or updated separately by the owner; it is never derived silently from the purchase rate.
- Existing sales keep their saved RateHistoryID and snapshots. New sales use the rate applicable to their Billing Date.

9. Damage Entry

Damage has two business outcomes. The user must choose the outcome instead of using one generic damaged quantity.

Damage option	Stock behavior	Profit behavior
REPAIRABLE	Move Saleable quantity into REPAIR HOLD at the same godown. Total owned physical stock is unchanged; Available Stock falls.	No loss yet. Optional repair charge posts as an operating expense.
REPAIRED	Linked reversal moves no more than the outstanding quantity from REPAIR HOLD back to Saleable. Available Stock rises.	Only any entered repair charge remains.
UNREPAIRABLE	Write quantity out from Saleable or REPAIR HOLD. Physical and Available Stock reduce.	Post owner-approved loss cost once to STOCK OUT / IRREPARABLE DAMAGE expense.
ENTRY REVERSAL	Reverse an erroneous damage event through its DamageID; never delete the original row.	Reverse the linked expense once when applicable.

- Required fields: Box image/code, Godown, quantity, date, repairability, reason and mandatory remarks. Raw Materials do not use Damage Entry because their physical stock is not tracked.
- Damage quantity cannot exceed eligible unreserved Saleable or Repair Hold quantity. Reserved damage requires a linked ReservationID and manager decision to replace/release it.
- For unrecoverable damage, show the suggested cost effective on the Damage Date but require Owner/authorized Unit Loss Cost or Total Loss Cost; store both the rate basis and manual override audit.
- Damage expense is an operating stock-loss expense, displayed under the requested STOCK OUT heading and deducted once in Net Profit. It is not mixed into fixed overhead or order direct cost.
- A repair action must link the original DamageID and cannot repair more than its outstanding repairable balance.

10. Transfer Entry

1. Select Box, Source Godown, Destination Godown and quantity. Optional tracked Gift Bags may also transfer as PCS; Raw Materials do not use Transfer Entry.
2. Source and Destination must be different; quantity must be positive and within source available stock.
3. Post TRANSFER_OUT and TRANSFER_IN under one TransferID atomically.
4. Total company physical stock does not change; only location balances change.
5. Transfer cancellation posts the exact reverse pair; rows are never deleted.

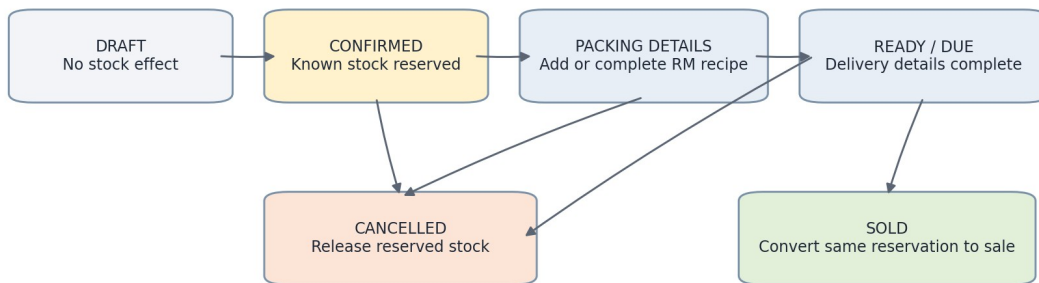
11. Other Stock Out

Use for personal use, promotion, donation, free sample, home use or another non-sale consumption. Reason, item quantities and remarks are mandatory. Physical Box quantity is deducted when a box is issued. Raw-material components create calculated Usage and a purchase-rate expense snapshot, but no raw-material stock movement because raw-material balances are not maintained. The purchase-based cost posts once to STOCK OUT / OPERATING EXPENSE for Net Profit.

PART C - RESERVATION, SALES AND COSTING

12. Reservation workflow

Reservation lifecycle - no duplicate stock deduction



A reservation is a customer commitment before final billing/delivery. It stores customer, delivery and quoted-price details while reserving physical Box Stock where a box is included. Raw-material lines are recipe/packing requirements only. Converting a reservation to a sale uses the same ReservationID link and must not deduct Box Stock twice.

12.1 Reservation header

- ReservationID, Reservation Date, Customer, optional Phone and optional company name.
- Promised Delivery Date/Time, pickup/delivery mode, delivery address and contact person.
- Finalised selling rate/amount, advance amount/payment status if used, remarks and owner approval.
- Status: DRAFT, CONFIRMED, MATERIALS_PENDING, READY, PARTIALLY_SOLD, SOLD, CANCELLED or EXPIRED.
- Multiple box/set lines are allowed under one reservation.

12.2 Reservation modes

Mode	Stock reserved now	Later action
BOX RESERVATION ONLY	Box quantity	Add raw-material packing details later if required
FULL RESERVATION	Box quantity only	Store specified raw-material recipe without reserving RM stock; prepare and deliver
ITEMS COMBO RESERVATION	None	Store set recipe/quantities; convert to sale on delivery

12.3 Reservation rules

- Confirmation immediately reduces Available Stock through RESERVE movements; Physical Stock is unchanged.
- If raw materials are not yet specified, do not guess them. The reservation is flagged MATERIALS_PENDING, meaning packing details are incomplete; it is not a raw-material stock reservation.

- MATERIALS_PENDING stores a Material Details Due Date/alert linked to the promised Delivery Date so staff complete the recipe before preparation.
- When packing details are entered, an optional Box Default Recipe may prefill common materials. Staff may add any other active material through Product Group -> Raw Material and must confirm Actual Packing Quantity before READY.
- Before READY/SOLD, each raw-material line must have a valid unit, Actual Packing Quantity and applicable rates. The system cannot block for insufficient raw-material stock because that stock is not tracked.
- Quoted selling price is stored and remains unchanged unless an authorized user edits it.
- Store quote-date cost snapshot for quotation analysis. Final accounting GP uses Billing Date rates/issued cost when converted to sale.
- Reservation cancellation releases reserved Box quantities; it does not add to Physical Stock because Physical was never reduced. No raw-material stock is released.
- A partial conversion creates an Order for only the delivered quantities, reduces physical/reserved Box Stock by that same amount, and leaves the remaining Box quantity reserved. For a no-box combo, only the remaining committed set quantity changes; no stock balance is involved.
- Delivery reminders and overdue reservation lists must be available by promised date.

13. Direct Sales Entry

One Sales Order has a header and one or more lines. A customer may buy multiple box designs and/or combo sets under the same OrderID. Each line stores its own sale type, quantity, rate, box, items and Gift Bag choice.

13.1 Mandatory Sales header

- Billing Date (mandatory), CustomerID/Name, optional Phone, OrderID, status and source (DIRECT or RESERVATION).
- Billing Date determines item eligibility and effective cost rates.
- Total Order Revenue is calculated in the backend; user enters rate and quantity, not the total.
- If GST is captured, show whether the entered customer rate is TAX-EXCLUSIVE or TAX-INCLUSIVE. Normalize it to taxable revenue and output GST explicitly; never infer or mix tax into profit.
- Tally/reference number may be added later without changing stock or cost snapshots.

13.1.1 Customer select-or-create

1. Typing Customer Name searches existing customers by normalized name, optional phone and company; show enough detail to choose the correct person.
2. User selects an existing CustomerID or taps CREATE NEW. Name is required; phone is optional and may remain blank.
3. Saving a sale/reservation atomically saves a new Customer record when CREATE NEW was selected, then links that CustomerID to the document.
4. Possible duplicate names/phones show a warning, not an automatic merge. Only the owner can merge duplicates, with all history relinked and audited.
5. The same customer becomes searchable on future entries and has a Customer Data/History view for orders, reservations, delivery details and totals.

13.2 Three Sale Types

Sale type	Box required	Raw materials	Quantity meaning	Inventory effect
BOX ONLY	Yes	Hidden / none	Number of boxes	Deduct boxes; apply Gift Bag cost and optional bag stock if enabled
BOX WITH ITEMS	Yes	Required per box	Number of boxes	Deduct boxes; calculate RM usage = item qty x boxes
ITEMS COMBO - NO BOX	No	Required per set	Number of sets	Calculate RM usage = item qty x sets; boxes sold = 0

UI LABELS: Use NO. OF BOXES / SETS and SELLING RATE / BOX / SET. For ITEMS COMBO - NO BOX, clear/disable Box selection and show ITEMS USED PER SET.

13.3 Line calculations

Total Required Item Quantity = Quantity per Box/Set x Number of Boxes/Sets

Line Product Revenue (excluding GST) = Taxable Selling Rate per Box/Set x Number of Boxes/Sets

Order Revenue for Profit = Sum of active Line Product Revenue + Packaging Amount Recovered + other customer charges, excluding output GST

- For a tax-inclusive entered rate, derive taxable rate and output GST using the configured tax rate and store both snapshots. Output GST is reported separately and is never Gross Profit.
- KG quantities accept decimals such as 0.250, 0.500 and 1.250.
- PCS quantities must be whole numbers.
- No active box with Available Stock = 0 appears in new sales selection, even if Physical Stock is positive but fully reserved.
- Existing historical orders continue to display inactive/out-of-stock boxes through saved BoxID and image snapshot.

13.4 Product Group and Raw Material selection in an entry

- 1. Select Box or NO BOX.** For a Box, prefill its active Default Recipe when one exists. Every active Raw Material still remains available to add.
- 2. Tap ADD RAW MATERIAL.** First select/search Product Group; then show only active Raw Materials in that group for the Billing/Calculation Date.
- 3. Select the exact material.** Store RawMaterialID, not only the Product Group. Display its Unit and applicable default quantity.
- 4. Confirm quantity.** Prefill from Box Recipe quantity, otherwise Material Default Selected Quantity, otherwise blank. Allow the user to overwrite it.
- 5. Save the working result.** For Sales/Reservation/Sample save Actual Packing Quantity. For Price Calculator use the selected/editable quantity only for the temporary result.

SELECTION EXAMPLE: User selects ASJ001, then ADD RAW MATERIAL -> Product Group CASHEW -> CASHEW W320. Default 0.250 KG appears; user changes it to 0.300 KG. For 4 boxes, Packing Sheet and Raw Material Usage show 1.200 KG. The permanent material default and Box Default Recipe remain unchanged.

13.5 Selected Quantity vs Actual Packing Quantity

Field	Rule
Selected Quantity	Optional searchable preset from the material/group and the primary quick input in Price Calculator. Saved workflows may use it only as a prefill shortcut; it is never stored or posted as the final quantity.
Actual Packing Quantity / Box or Set	Editable value prefilled from Selected Quantity or typed manually. Mandatory for saved sample/sale/reservation items and stored with the recipe/order version.
Total Required	Actual Packing Quantity x Number of Boxes/Sets. This is posted to the Raw Material Usage Ledger only, not a physical stock ledger.

- Actual Packing Quantity remains visible in item details, Sample Master, Sales Register expansion and the staff Packing Sheet. It must never be replaced by a later master default.
- Packing Sheet is searchable by OrderID, Customer, Delivery Date, BoxCode/Image or SampleID and shows per-box/set quantity plus total required for every item.
- A staff-friendly display may show both 0.250 KG and 250 g, while the database and calculations always use 0.250 KG.

14. Gift Bag and Packaging in Sales

1. User selects Bag Required = YES or NO for each box/set line.
2. NO clears/hides Bag Size and sets bag quantity/cost to zero.
3. YES shows enabled SMALL/MEDIUM/LARGE sizes; size is mandatory.
4. Bag Quantity equals line boxes/sets automatically.
5. Applicable rate is selected by Billing Date and stored as a cost snapshot.
6. Bag cost is included in both purchase-based and internal pricing cost, so it affects Actual GP and Commercial GP. It is never automatically added to a manually entered Selling Rate.
 - Store Packaging Cost Incurred (direct cost) and Packaging Amount Recovered/Received from customer as separate fields when applicable.
 - Packaging Recovery may be stored at order level or allocated to lines, but never both for the same amount. The Sales Register must show where it was recorded.
 - If the entered Selling Rate already includes packaging, Packaging Amount Recovered remains zero; never enter the same recovery twice.
 - Display Net Packaging Burden = Packaging Cost Incurred - Packaging Amount Recovered so the owner can see that recovery reduces the packaging burden.

- For accounting profit, Packaging Amount Recovered is included in customer/order revenue and the full Packaging Cost Incurred remains a direct cost. Never also subtract the recovery from cost in the same GP formula, which would count the benefit twice.
- Price Calculator, Sample Combo, quotation and Sales Register must show packaging cost, recovery and net burden separately when used.

Net Packaging Burden (display) = Packaging Cost Incurred - Packaging Amount Recovered

15. Automatic stock posting from Sales

Event	Box stock movement	Raw-material usage entry	Revenue/profit
Direct Save	DIRECT_SALE when a box exists	Create DIRECT_SALE_USAGE; no RM stock effect	Create snapshots and active report totals
Reservation -> Sold	RESERVATION_SOLD for reserved boxes	Create RESERVATION_SOLD_USAGE once	Reuse reservation, create OrderID, no double deduction
Edit/Update	Post net box-stock delta	Post/version net calculated usage delta	Replace current snapshot; retain version history
Cancel	SALE_CANCEL_REVERSAL for boxes	Reverse/exclude active usage once	Status CANCELLED; exclude revenue and GP

ATOMICITY: Order, item lines, cost snapshots, Box Stock movements, Raw Material Usage entries and reports must commit together. A failure leaves the entire prior state unchanged and preserves user-entered form data.

16. Sales Edit, Update and Cancel

- Edit loads the same OrderID and current active version; it never creates a new OrderID.
- Created At remains unchanged; Updated At and Version increment on successful update.
- Changing Billing Date rechecks eligible materials/boxes and recalculates applicable rates for the new date.
- If an existing selected material is not eligible on the new Billing Date, highlight it and block update until resolved. Never silently delete it.
- Update computes stock delta between old and new version. It must not reverse and re-deduct in a way visible to users if the transaction fails.
- Cancel requires confirmation/reason, reverses inventory once, retains the record with CANCELLED status, disables further ordinary edit/cancel, and removes it from active reports.
- Repeated Edit -> Update cycles must work without losing controls or producing duplicates.

17. Effective-date rate engine

Every rate table uses non-overlapping periods. The lookup key is ItemID + Billing Date. The rate screen remains owner-only and the Sales Register shows totals, not detailed backend rates.

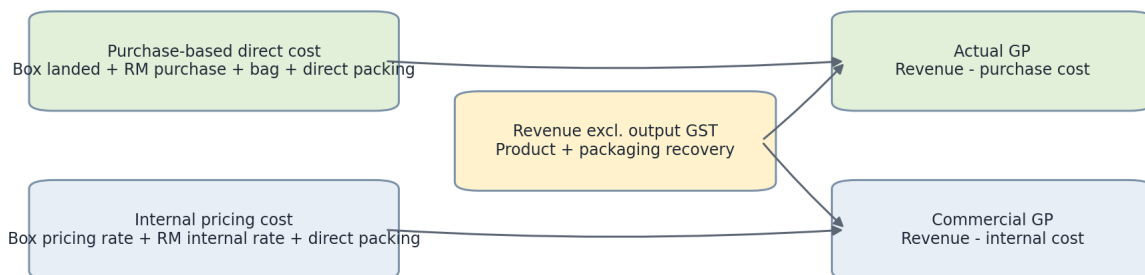
Action	Behavior
CREATE MATERIAL / BOX RATE	Enter rate and Effective From. Previous open period closes one day before the new start.

Action	Behavior
UPDATE RATE	Create a new rate period; never overwrite the prior period.
EDIT RATE	Correct a wrongly entered rate in the same period; dates remain locked for normal users.
ADMIN DATE CORRECTION	Owner-only exceptional workflow; prevent overlap; audit old/new dates and reason.
INACTIVATE	Set Inactive From; hide from new entries outside active period; keep all history.
REACTIVATE	Create a new active period for the same ItemID; do not create a duplicate material.

- Normal future rate updates do not change earlier sales.
- A historical rate correction identifies affected orders by RateHistoryID or ItemID + Billing Date, recalculates costs/GP, logs before/after values and leaves selling price unchanged.
- When accounting-period locking is introduced, a correction to a locked period creates an adjustment record instead of silently rewriting closed reports.
- Price Calculator and Sample Combo use their displayed Calculation Date, which defaults to today; saved Sales always uses Billing Date snapshots.

18. Cost and profit engine

Two cost views with one consistent revenue basis



REVENUE BASIS: Actual Sale Revenue used for GP is the customer charge excluding output GST. It includes separately charged Packaging Recovery once. Output GST remains a tax liability, not revenue or profit.

18.1 Purchase-based direct cost

Actual Direct Cost = Box Landed Cost + Raw Materials at Purchase / Actual Cost Rate + Gift Bag Cost + Other Direct Order Costs

Actual Gross Profit = Actual Sale Revenue - Actual Direct Cost

RAW-MATERIAL COST POLICY: Raw Materials are costed using the owner-provided Purchase / Actual Cost Rate effective on the Billing Date. FIFO and moving weighted average are not applicable because raw-material receipt quantities, batches and balances are not recorded. Each order stores the applied RateHistoryID, rate and amount for audit.

18.2 Internal pricing cost

Commercial Cost = Final Unit Box Rate + Raw Materials at Internal Selling/Costing Rate + Direct Packing/Gift Bag Cost

Commercial Gross Profit = Actual Sale Revenue - Commercial Cost

18.3 Margins and markup

Actual Gross Margin % = Actual Gross Profit / Revenue x 100

Commercial Gross Margin % = Commercial Gross Profit / Revenue x 100

Realized Markup % = (Revenue - Commercial Cost) / Commercial Cost x 100

If Revenue or Commercial Cost is zero, the related percentage displays N/A and is never divided by zero or shown as an infinite percentage.

The Purchase-vs-Internal-rate spread, box Special Extra Profit, normal markup and any difference between recommended and actual selling price are visible through the separation of Actual GP and Commercial GP. Actual GP is the primary management gross-profit view; Net Profit becomes the final profitability view only after overhead and operating stock-out expenses are deducted.

18.4 Overhead and net profit

Net Profit = Actual Gross Profit - Period Fixed Overhead - Operating Stock-Out / Damage / Sample Expenses

- Overhead includes godown rent, salary, electricity, software/admin and other fixed period expenses.
- Overhead is entered by month/date and category, then deducted once in the profit report.
- Overhead must never be added to box cost, raw-material rate, Gift Bag rate or order direct cost.
- Irreparable damage, free samples, home/personal use and similar stock-outs appear under Operating Expenses / STOCK OUT, not fixed overhead, and are deducted exactly once.

18.5 Test example - No-Box Combo

Input / result	Value
Sets and selling rate	2 sets x INR 800 = INR 1,600 revenue
Almond	0.250 KG/set; purchase INR 800/KG; internal INR 900/KG
Cashew	0.250 KG/set; purchase INR 900/KG; internal INR 1,050/KG
Total usage	0.500 KG Almond + 0.500 KG Cashew
Purchase cost per set	INR 425

Input / result	Value
Internal value per set	INR 487.50
Actual GP	INR 1,600 - INR 850 = INR 750
Commercial GP	INR 1,600 - INR 975 = INR 625
Dashboard quantities	Boxes Sold = 0; Combo Sets Sold = 2

PART D - PRICE CALCULATOR AND SAMPLE COMBO

19. Price Calculator

Price Calculator is a temporary quotation tool. It does not create an order, reserve stock or deduct inventory.

- Calculation Date defaults to today and controls the effective box, raw-material and Gift Bag pricing rates shown in the quotation.
- Show only active, positive-available-stock boxes as image cards.
- Support BOX ONLY and BOX WITH ITEMS. ITEMS COMBO - NO BOX is entered directly as an actual sale and is not needed here.
- For BOX WITH ITEMS, selecting a Box prefills its optional Default Recipe. The user may remove a line or add any active Raw Material through Product Group -> Raw Material selection.
- The calculator quantity priority is Box Recipe quantity -> Material Default Selected Quantity -> blank manual entry. It remains editable and is used only for the current price calculation; the calculator does not create Actual Packing Quantity, a recipe or a transaction unless the user later enters/saves a real workflow.
- Use Final Unit Box Rate from Box Cost History. Special Extra Profit is already built into that rate and is not a separate final-column calculation.
- Use Internal Selling/Costing Rates applicable to the Calculation Date for materials.
- Apply the normal markup after the complete pricing base is calculated.
- Clearing the calculator has no backend transaction effect.
- Show Packaging Cost, optional Packaging Recovery and Net Packaging Burden without double-counting recovery in profit.

Pricing Base = Final Unit Box Rate + Raw-material Internal Value + Gift Bag Cost + Other Direct Packing Charge

Recommended Selling Price = Pricing Base x (1 + Normal Markup %)

20. Sample Combo and Sample Master

Sample Combo is a saved recipe/costing and packaging tool. Saving/editing a recipe or changing its operational Units Made counter does not deduct Box Inventory or create Raw Material Usage. The recipe stores the Actual Packing Quantity that will truly go into each box/set; its displayed price uses rates applicable to its displayed Calculation Date, which defaults to today.

20.1 Sample recipe fields

- SampleID, Sample Name/No., BoxID/Image or NO BOX, recipe items, Actual Packing Quantities, units, Units Made and status.
- Calculated price from the Calculation-Date Final Unit Box Rate + internal raw-material rates + Gift Bag/direct packing + normal markup.
- Manual MRP field in Sample Master.
- Edit reloads the same SampleID; update changes recipe version; cancel preserves history and hides from new use.
- Inactive/out-of-stock boxes cannot be selected for a new recipe but existing saved samples remain readable.
- Selecting a Box may prefill its optional Default Recipe. The owner can add any other active Raw Material through Product Group -> Raw Material; saving the Sample stores its own versioned recipe and does not silently change the Box Default Recipe.
- Selected Quantity may prefill a row, but only the saved Actual Packing Quantity is used when the sample is sold or issued.

20.2 Units Made and inventory actions

- Units Made is an editable operational counter for how many units of the same Sample Combo are prepared/planned. It does not change Physical, Reserved or Available Stock by itself, per the owner-approved rule.
- Units Available = Units Made - Units Transferred to Sale - Units Issued Free/Home. Every adjustment is versioned and searchable.
- Transfer to Sales asks for customer, units, Billing Date, rate and optional additional lines. It creates/links a normal OrderID, uses Actual Packing Quantity, decreases Units Available, posts Box Stock when applicable and creates calculated Raw Material Usage once.
- If transfer quantity exceeds recorded Units Available, warn clearly and require confirmation to add the shortage as newly made before transfer. Never leave a negative sample-unit balance.
- A Sample Master search shows recipe, actual per-unit packing quantities, Units Made/Available, transferred units and issue history at all times.

Action	Inventory effect	Use case
SAVE RECIPE	None	Design and price a sample/combo
UPDATE UNITS MADE	None; operational counter only	Record/edit quantity prepared or planned
TRANSFER TO SALES	Create sale; deduct Box if used; calculate RM usage	Sell one or more sample units to a customer
GIVE / HOME / PERSONAL USE	Deduct Box if used; calculate RM usage once	No revenue; component cost becomes STOCK OUT expense
CANCEL RECIPE	No stock effect	Preserve history and hide from new selection

DOUBLE-DEDUCTION CONTROL: Sample recipe/Units Made changes post nothing. Final disposition posts exactly once: Sales creates Box movement when applicable plus Raw Material Usage; Give/Home/Personal Use creates the applicable Box stock-out, calculated Raw Material Usage and expense. Repeated Save/Transfer retries must be idempotent.

PART E - REGISTERS, REPORTS AND USER EXPERIENCE

21. Sales Register

Sales Register is the audit-facing list of all active and cancelled sales. Use a compact summary with expandable line/item details rather than showing every raw-material rate in the main table.

Summary field	Behavior
Order ID / Billing Date	Permanent identity and mandatory business date
Customer / Phone	Searchable; links to Customer History
Order status	ACTIVE/POSTED/CANCELLED; cancelled remains visible
Sale lines	BOX/SAMPLE indicator, box image/code or NO BOX, sample/reservation source, sale type, box/set quantity, selling rate and line revenue
Financial totals	Order Revenue, Commercial GP, Actual GP and margins; owner/authorized roles only
Items	Expandable Actual Packing Quantity per box/set plus total required; rates remain backend-only; open Packing Sheet action
Actions	Edit, Update, Cancel and Print/Export according to role/status; reservation conversion is performed from the Reservation workflow

22. Inventory reports

- Box Stock by Godown: Saleable, Repair Hold, Total Physical, Reserved, Available, Written Off and stock value. Optional tracked Gift Bags appear separately.
- Low Stock / Out of Stock: active designs below threshold; restock link opens Purchase Entry.
- Movement Register: BOX/SAMPLE source indicator, image/code or SampleID, document, type, date, source/destination, quantity, before/after and user.
- Reservation Stock: customer, delivery date, reserved boxes, pending packing details and age. Raw Materials are shown as recipe requirements, not reserved stock.
- Damage Register: repairable/repaired/unrepairable status, outstanding Repair Hold, item, godown, quantity, linked reversal, cost impact, remarks and approver.
- Box Purchase Register: supplier, accepted/damaged box quantity, landed rate, transport and cost-period effect.

23. Raw Material Usage

- Monthly/yearly calculated consumption by raw material and unit; Usage is not Physical, Reserved or Available Raw Material Stock.
- Include ACTIVE BOX WITH ITEMS and ITEMS COMBO - NO BOX sales.
- Include/identify Sample-to-Sale usage and free/home/personal stock-out consumption separately.

- Cancelled sales and cancelled item-line versions contribute zero.
- Drill down from total usage to OrderID/SampleDispositionID/StockOutExpenseID.
- Use for purchasing forecasts and year-over-year comparisons.
- Do not display a current raw-material balance, shortage or FIFO valuation unless a future quantity-based Raw Material Inventory module is separately approved and implemented.

24. Profit dashboard

Metric	Monthly / yearly definition
Revenue	Sum active order revenue excluding output GST by Billing Date; include packaging recovery/customer charges once
Active Orders	Distinct active OrderIDs
Boxes Sold	Sum quantities for BOX ONLY + BOX WITH ITEMS
Combo Sets Sold	Sum quantities for ITEMS COMBO - NO BOX
Commercial GP / Margin	Revenue less internal pricing cost
Actual GP / Margin	Revenue less purchase-based direct cost
Fixed Overhead	Godown rent, salaries, electricity, software/admin and other period costs, deducted once
Stock-Out Operating Expense	Irreparable damage, free sample, home/personal use and other non-sale cost, deducted once
Net Profit / Margin	Actual GP less fixed overhead and stock-out operating expenses
Packaging	Cost incurred, amount recovered and net packaging burden, without double counting

All report queries must read the transaction/master ledgers, not rendered Sales Register screen columns. Filters: year, month, date range, godown, customer, sale type, box, material and status.

25. Other operational views

- Customer Data/History: searchable customer master, orders, reservations, quantities, prices, delivery details, status and lifetime revenue/profit; phone remains optional.
- Packing Sheet: mobile/print view by OrderID/Customer/Delivery/Sample showing Box Image, box/set count, Actual Packing Quantity and Total Required for every material.
- Delivery Board: today/upcoming/overdue reservations; pickup/delivery details and readiness.
- Rate History: item, rate type, start/end date, current flag, update/edit action and audit trail.
- Sample Master: image, recipe, Calculation-Date calculated rate, MRP, status, Units Made, Transfer to Sale and Give/Home/Personal Use actions.
- Owner Dashboard: low stock, pending reservations, today deliveries, sales and GP snapshots.

26. Application navigation

Menu	Main screens
Dashboard	Owner summary, alerts and shortcuts
Inventory	Stock, Purchase, Damage, Transfer, Other Stock Out, Movement Register
Reservations	New, Pending Materials, Ready, Due, Sold, Cancelled
Sales	New Sale, Sales Register, Customer History
Costing	Box Cost, Raw Materials, Rate History, Gift Bag, Business Settings
Pricing & Samples	Price Calculator, Sample Combo, Sample Master, Units Made and Sample Disposition
Reports	Inventory, usage, purchases, revenue, profit, overhead and exports
Administration	Users, roles, godowns, masters, audit, import/export and backups

PART F - DATABASE AND TECHNICAL DELIVERY

27. Required database entities

Names may follow the developer's convention, but the relationships and responsibilities below are mandatory.

Domain	Tables / entities	Purpose
Identity	users, roles, permissions	Authentication and field/action security
Masters	boxes, box_images, godowns, product_groups, raw_materials, quantity_presets, box_recipe_templates, box_recipe_items, gift_bags, customers, suppliers, settings	Permanent business identity, optional Box defaults and configuration
Rates	box_rate_history, raw_material_rate_history, gift_bag_rate_history	Effective periods, rate type, applied dates and audit
Purchases	box_purchase_headers, box_purchase_lines, purchase_cost_allocations	Box supplier entry, landed costs and stock-in source
Inventory	box_stock_movements, stock_reservation_links, damage_events, damage_repairs, box_stock_balance_view	Box movement ledger, condition buckets and derived balances
Raw Material Usage	raw_material_usage_entries, usage_reversals	Calculated consumption from sales/sample disposition; no physical RM balance
Reservations	reservation_headers, reservation_lines, reservation_items, delivery_details	Customer commitment, reserved Box Stock and non-stock RM packing requirements
Sales	order_headers, order_lines, order_items, order_cost_snapshots, order_versions	Revenue, quantities, applied rates and audit
Samples	sample_recipes, sample_recipe_items, sample_versions, sample_unit_events, sample_dispositions	Recipe, Units Made and final disposition
Expenses	overhead_entries, overhead_categories, stock_out_expenses, packaging_recoveries	Fixed overhead, operating losses and packaging recovery
Audit	audit_events, reversals, posting_idempotency	Before/after values, user, reason and retry safety

28. Key entity relationships

- OrderHeader 1-to-many OrderLines; each line 0/1 BoxID and many OrderItems.
- ReservationHeader 1-to-many ReservationLines/Items; conversion stores resulting OrderID.
- Every Box Purchase/Sale/Reservation/Transfer/Damage/final Sample Disposition that affects a physical box posts one or more BoxStockMovement rows through DocumentID.
- Every sale or final Sample Disposition containing Raw Materials posts RawMaterialUsageEntry rows through the same DocumentID; these rows never change a physical RM balance.
- Every Sample Transfer to Sale stores SampleID + OrderID. Every free/home disposition stores SampleID + StockOutExpenseID.
- Every costed line stores RateHistoryID + AppliedRate + AppliedQuantity + AppliedAmount.

- ProductGroup 1-to-many RawMaterials. A transaction item always stores the exact RawMaterialID; ProductGroupID supports selection and reporting only.
- Box 0/1-to-many versioned Default Recipe Items. The template may prefill a working document, but every active RawMaterialID remains eligible for every Box and an order/sample stores its own copied item snapshot.
- When Gift Bag Track Inventory is enabled, GiftBagID participates in the same generic StockMovement and availability rules as a PCS item; when disabled, no Gift Bag stock rows are created.
- BoxImage is referenced by BoxID; transaction image snapshots are optional display backups, never the identity.
- OrderVersion preserves prior business values; only one current active version is reportable.

29. Data types and constraints

Data	Requirement
Money	Decimal/Numeric, never binary floating point; minimum 2 decimal precision
KG quantity	Decimal, minimum 3 decimal places
PCS/boxes/sets	Positive integer
Dates	Business date stored separately from created timestamp; display DD/MM/YYYY; timezone Asia/Kolkata
IDs	UUID or safe sequence; unique constraints on BoxCode and active material display identity
Effective periods	No overlap per ItemID + RateType; End Date derived from next Start Date
Packing quantity	Prefill priority: Box Recipe quantity, then Material Default Selected Quantity, then blank; saved Actual Packing Quantity mandatory for posted item lines
GST	Taxable value, GST rate, output GST and tax-inclusive/exclusive basis stored separately; GST is never inferred from a total
Box stock	No negative Saleable, Repair Hold, Physical, Reserved or Available balance after committed transaction; no equivalent RM balance exists
Status	Enumerated state machine; invalid transitions rejected

30. Transaction, concurrency and performance

- Use database transactions and row-level locking/version checks when posting Box Stock.
- Recheck availability inside the transaction immediately before commit; UI display alone is insufficient.
- Use idempotency key per Save/Update/Convert action to prevent duplicate posting after network retry.
- Batch line/item inserts and calculations; avoid one query per line.
- Index business dates, statuses, ItemIDs, BoxIDs, GodownIDs, OrderIDs, ReservationIDs and RateHistoryIDs.
- Maintain report summaries or optimized views after successful posting; never scan all history on every dashboard open.

- All failures return a user-readable message and preserve the draft form.

31. Audit requirements

Event	Audit data
Create/Update/Cancel	User, timestamp, source device, old version, new version, reason
Rate correction	Old/new rate, period, impacted documents and before/after GP
Stock correction	Movement reversal/adjustment, before/after balances and approver
Master inactive/reactivate	Old/new status, effective date and user
Permission change	Administrator, affected user/role and before/after permissions

32. Validation and error messages

- No entry without mandatory business date.
- Box/material must be eligible for that business date and transaction type.
- Quantity and unit rules enforced server-side.
- No duplicate BoxCode, overlapping rate period, negative Box Stock or invalid reservation transition.
- Missing applicable rate blocks Save/Update and names every missing item/rate type.
- Changing a date or line shows the user what will be repriced or released before confirmation.
- A failed action changes nothing in backend and leaves user input intact.

33. Security, privacy, backups and ownership

- Individual password-protected accounts; passwords stored only as strong salted hashes. Owner MFA, role-based access, session revocation and login/audit history are mandatory.
- Encrypt data in transit and at rest. Owner-only screens protect purchase rates, internal rates, GP, overhead, exports, users and corrections.
- Automated encrypted backups with point-in-time recovery where supported, daily backup, separate copy and tested restore procedure.
- Recycle Bin/soft delete allows the owner to restore accidentally removed eligible drafts/masters with all links. Production transactions use Cancel/Reverse, not hard delete.
- A dedicated REMOVE TEST DATA / PRODUCTION RESET deletes only TEST-environment documents after encrypted backup and typed owner confirmation; it preserves schema, users, configuration and acceptance evidence.
- Owner can export Box Master, stock, movement, rate history, orders, items, reservations, purchases and reports to standard files.
- Source code, database schema/migrations, deployment scripts and documentation belong to ASJ and are stored in an ASJ-controlled repository/account.
- No production secret or credential is hard-coded in source control.
- Staging/test and production environments are separated.

- No panic-delete, hidden-data or unauthorized-access feature may be built. Retention and lawful disclosure rules are configured with ASJ's CA/legal adviser before go-live.

GST RECORD SAFETY: Security prevents unauthorized access; it cannot be used to conceal or destroy records required by law. Current CGST Section 36 states a 72-month retention period from the annual-return due date, with longer retention for specified proceedings/investigations. Keep this configurable and have ASJ's CA confirm the current rule at go-live.

Official references: [CBIC CGST Act - Accounts and Records](#) | [CBIC Accounts and Records Rules](#)

33.1 Mobile parity and recovery behavior

- Every authorized daily and owner workflow available on laptop must also work on mobile: masters, images/camera upload, purchase, damage/repair, transfer, reservation, sale, edit/update/cancel, samples, packing, reports, export and restore.
- Use responsive forms, image cards, search, large touch controls and one-column mobile layouts; no hover-only or custom-menu-only action.
- Interrupted connections preserve a draft and use idempotency on retry. The user sees success only after the complete database transaction commits.
- Restore is owner-initiated, not silent: selecting a recycle-bin item or recovery point restores all linked data atomically and records who restored it. A successful intentional permanent TEST reset does not auto-return unexpectedly.
- Acceptance testing must include the owner's iPhone and at least one Android/desktop browser at practical screen sizes.

34. Developer deliverables

1. Approved screen map and clickable prototype.
2. Relational schema and migration scripts.
3. Responsive owner/staff application with role-based access.
4. Tested APIs/domain services for inventory, reservation, sale, costing and reporting.
5. Import templates/tools for owner-supplied Box/Raw Material/Product Group masters, Raw Material defaults, optional Box Default Recipes, box images/codes, Box opening stock by godown, box costs and Raw Material rates; no RM stock import and no dependency on the incomplete Costing Sheet.
6. Automated unit/integration/end-to-end test suite.
7. Audit log, backup/restore and data export.
8. Mobile parity report proving every authorized workflow has been tested on phone and desktop.
9. Deployment documentation, source repository handover and owner training.
10. Production cutover checklist and signed reconciliation report.

35. Scope sequencing

Phase	Must complete
1. Foundation	Users/roles, masters, permanent IDs, images, rate history, opening import
2. Box Inventory	Box purchase, balances, damage, transfer, other out, movement/audit
3. Reservations	Box-only/full reservation, delivery details, cancel and conversion
4. Sales & costing	Three sale types, multi-line orders, Gift Bag, stock posting, edit/cancel, snapshots
5. Pricing & samples	Price Calculator, Sample recipes, Units Made and final disposition
6. Reports	Sales, stock, usage, profit, overhead, customer and export
7. Production	UAT, opening sign-off, deployment, training, support

35.1 Five-day MVP build sprint - ambitious target

Day	Integrated target
1	Schema, users/roles, Box/Raw Material/Product Group/Customer masters, quantity presets, optional Box Default Recipes, Box opening import and rate history
2	Box purchase/balances, repairable/unrepairable damage, repair/reversal, transfer and stock-out; RM usage ledger structure
3	Reservations, three sales types, multi-line orders, customers, packaging, costing, Edit/Update/Cancel and stock posting
4	Price Calculator, Sample Master/units/disposition, Packing Sheet, usage/profit/inventory reports and mobile layouts
5	Core end-to-end regression, owner mobile/desktop UAT, permissions, retry checks, backup/restore proof, defect list and go/no-go decision

SPRINT BOUNDARY: Five days targets an integrated MVP with temporary data, not automatic production certification. Security hardening, unresolved defects, performance work, restore testing and owner acceptance continue until every production-readiness gate passes. The calendar never overrides a failed test.

INTEGRATION RULE: Do not mark a module complete only because its form saves. After the final code change, rerun the complete chain from masters/box purchase through Box Stock, reservation, sale/sample disposition, RM usage, update/cancel, customer history, profit, backup and restore.

PART G - ACCEPTANCE TESTS

36. Minimum acceptance matrix

Test	Pass condition
Opening stock	Import approved Box balances by BoxID/godown; location and company totals reconcile exactly; no Raw Material stock balance is requested or created
Product Group creation	Create CASHEW, then W320 and W240 under it; selection saves exact RawMaterialID and group reports aggregate both variants
Group-filtered selection	Selecting CASHEW shows only active Cashew materials for the transaction date; inactive group/material remains in history but is hidden from new entry
All-material box eligibility	Any active Raw Material can be added to any active Box; no hidden Box-to-Material restriction blocks selection
Box Default Recipe	Selecting ASJ001 prefills its active recipe; adding RAISIN or overriding quantity changes the working document only and does not modify the permanent template without owner Save/Update Default Recipe
New design purchase	Unique code/image created, cost period created, stock added to selected godown
Existing restock	Image selection reuses BoxID; no duplicate Box Master row
Purchase costing	Transport and Special Extra Profit calculations match formulas; overhead excluded
Receipt rejection	Supplier-returned quantity never enters stock or expense; accepted-but-damaged quantity follows the selected damage outcome
GST/ITC treatment	Recoverable input GST is excluded from landed cost; non-recoverable GST is included; output GST is excluded from revenue/GP
Tax-inclusive sale	Entered inclusive amount splits into taxable revenue and output GST exactly; profit uses taxable revenue only
Repairable damage	Saleable -> Repair Hold; available falls; physical total remains; linked repair restores no more than outstanding
Irreparable damage	Stock writes out; owner cost posts once to STOCK OUT expense; reversal reverses stock and expense once
Transfer	Source decreases, destination increases, company total unchanged
Reserve boxes	Available decreases, physical unchanged; delivery and finalised price saved
Full reservation	Box quantity reserves; specified Raw Materials save as packing requirements only; pending-details option works when recipe is unknown
Reservation sale	Same reservation converts to order and deducts physical/reserved once
Partial reservation sale	Delivered quantity sells once; only the remaining quantity stays reserved; later sale/cancel reconciles to zero

Test	Pass condition
Reservation cancel	Reserved releases; physical unchanged; history remains
BOX ONLY sale	Box stock deducts; items hidden; revenue/boxes sold correct
BOX WITH ITEMS sale	Box deducts; Raw Material Usage equals actual item qty x box quantity; no RM stock balance changes
NO-BOX combo sale	No box deduction; RM Usage equals actual item qty x sets; Combo Sets metric increments
Quantity default/override	Priority is Box Recipe quantity -> Material Default Selected Quantity -> blank; manual override saves Actual Packing Quantity; Packing Sheet and usage use actual value
KG/PCS	0.250 KG x INR/KG calculates directly; PCS rejects decimals and uses INR/PCS
Gift Bag	NO hides size; YES uses enabled size; qty follows units; cost affects both GP views once; optional bag stock posts only when tracking is enabled
Date eligibility	Future-created material is not selectable for an earlier Billing Date
Rate update	New period closes prior period; past sale unchanged
Rate correction	Affected costs/GP recalculate with full audit; selling price unchanged
Sales update	Same OrderID, version increments, only stock delta posts, no duplicate
Sales cancel	Stock reverses once; reports exclude; cancelled record remains
Sample recipe	Save/edit changes no stock; displayed price follows rates effective on the selected Calculation Date
Sample units	Recipe/Units Made edit posts no stock and never creates negative Units Available
Sample to sale	Uses Actual Packing Quantity; creates/links order; Box deducts if used; RM Usage posts once; sample units decrease
Sample give/home	No revenue; Box deducts if used; RM Usage and purchase-rate cost post once to STOCK OUT expense
Packaging recovery	Cost, recovery and net burden display; GP counts the recovery benefit exactly once
Zero denominator	Zero revenue/cost displays N/A for the affected margin/markup and never errors or shows infinity
Customer create/select	Name required, phone optional; existing search/new save works; history links by CustomerID
Reports	Revenue, orders, boxes, combo sets, GP, usage and net profit reconcile to ledgers
Concurrency	Two users cannot oversell or over-reserve the same stock
Retry safety	Repeated API/request retry creates no duplicate document or movement
Mobile parity	Every owner-authorized operation passes on phone and desktop without a desktop-only control

Test	Pass condition
Recovery	Eligible accidental deletion restores all linked records; test reset cannot remove production records

37. Production readiness gate

- All acceptance tests pass with temporary data.
- Every entry type is retested together after the last change; isolated earlier passes are not sufficient.
- Owner verifies every major workflow on mobile and desktop.
- Temporary orders, samples, materials and movements are removed through an approved test reset, not mixed with production.
- Opening Box Stock by godown, optional tracked Gift Bag stock, box costs and Raw Material effective rates are signed off. No Raw Material stock quantity is required.
- Backups and restore are tested.
- Permissions are tested using Owner, Manager and Staff accounts.
- Developer hands over source, deployment access, schema, credentials ownership and documentation.

38. Deferred enhancements - design for, do not block core

- Tally integration, invoice import/export and payment reconciliation.
- Profit Planner: monthly/yearly target revenue, markup-to-margin conversion, break-even and target net profit.
- Purchase forecasting based on raw-material usage and seasonal sales.
- Barcode/QR scanning and printed labels.
- Customer quotations, WhatsApp notifications and delivery route integration.
- Accounting period lock and adjustment journals.
- Finished-goods/sample inventory if ASJ later pre-builds standard combos in quantity.

APPENDIX A - STATUS GLOSSARY

Object	Statuses
Master	ACTIVE, INACTIVE
Stock document	DRAFT, POSTED, CANCELLED / REVERSED
Reservation	DRAFT, CONFIRMED, MATERIALS_PENDING, READY, PARTIALLY_SOLD, SOLD, CANCELLED, EXPIRED
Sales Order	DRAFT, POSTED/ACTIVE, CANCELLED
Sample Recipe	ACTIVE, INACTIVE/CANCELLED
Sample Disposition	DRAFT, TRANSFERRED_TO_SALE, STOCKED_OUT, CANCELLED/REVERSED
Rate period	CURRENT, CLOSED, CORRECTED

APPENDIX B - OWNER DATA REQUIRED FOR GO-LIVE

Data	Owner provides	Developer validates/imports
Boxes	Image, final BoxCode, opening quantity at Home/Shop, initial unit cost	Unique ID/code, image storage, godown totals and opening movements
Raw materials	Product Groups; exact variants/names; KG/PCS; optional default/preset quantities; active dates; two rates; optional Box Default Recipes; no opening stock	Group-to-material links, all-material box eligibility, recipe versioning, quantity priority, non-overlapping rates, active/inactive history and non-stock policy
Gift Bags	SMALL/MEDIUM/LARGE cost rates, enabled flags and, if tracked, opening PCS by godown	Effective history, sales eligibility and optional stock movements
Business settings	Markup, packing rules, default tax/cost choices	Configuration and calculation tests
Customers	Names and available phone/address/delivery details; phone may be blank	Duplicate warnings, permanent IDs and history links
Packaging	Direct packaging costs and any separately recovered amount	Cost/recovery separation and no double counting
Overhead	Monthly categories and amounts when ready	Separate expense module; never direct product cost
Users	Names, phones/emails and role	Secure accounts and permission tests

APPENDIX C - FINAL IMPLEMENTATION CHECK

THE ONE-ENTRY TEST: For every business event ask: 'If the user enters this once, do stock, reserved quantity, costing, sales history and reports all update together without another manual entry?' If the answer is no, the workflow is incomplete.

THE HISTORY TEST: If a box/material becomes inactive, a rate changes, an order is updated or cancelled, can the system still explain the exact old transaction? If the answer is no, the data model is unsafe.

THE PROFIT TEST: Can Actual GP be reconciled to purchase-based direct costs while Commercial GP remains a separate pricing view and overhead is deducted once? If the answer is no, the profit model is wrong.

THE PACKING TEST: Can staff search an order/sample and see the exact Actual Packing Quantity per box/set and total required, regardless of the original preset or later master changes? If the answer is no, packaging control is incomplete.

THE RAW-MATERIAL SELECTION TEST: Can a user select Product Group -> exact active Raw Material, receive the correct default quantity, overwrite it, and add that material to any Box without silently changing the Box Default Recipe? If the answer is no, the product/recipe design is incomplete.